

DHRUV PANCHAGATTI

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EDUCATION

University of California, Irvine (UCI) Expected Graduation: [June 2029]

B.S. Electrical Engineering

Northwood High School Irvine, CA Expected Graduation: [June 2026]

GPA: 4.36 (Weighted) | Charter Oak Award Recipient

TECHNICAL SKILLS

Programming: Python, C++, Java

Engineering: Embedded Systems, Microcontrollers, Circuit Design, PWM Motor Control

AI / ML: Machine Learning, Data Analysis, RAG Systems, Prompt Engineering

Tools: Arduino, ESP32, CAD (3D Modeling and Printing), Soldering, Circuitry

EXPERIENCE

Founding Instructor — SciConnect Initiative (Nonprofit) [June 2024] – Present

- Co-founded and scaled a nationwide educational platform delivering Science Olympiad training across **36+ states**
- Created **200+ hours** of instructional video content, reaching **11,000+ students**
- Designed structured course pipelines to improve student performance and engagement at scale
- Contributed to growth strategy and learner conversion across digital platforms

AI Intern — GPT-Trainer / Petal / Go2Harvard Remote | [June 2024] – [Sept 2024]

- Published **8+** technical blog posts and analyzed **500+** data points to improve SEO performance
- Enhanced AI chatbot outputs through prompt engineering and RAG-based system optimization
- Collaborated with team to improve product functionality and user experience

PROJECTS

Gesture-Controlled / Embedded Robotics System + Published Research Paper

- Designed and built a robotics system integrating microcontrollers, motor drivers, and real-time control logic
- Implemented PWM-based motor control and hardware interfacing for precise movement
- Developed system architecture combining embedded hardware and software

Machine Learning Sports Analytics Model + Published Research Paper

- Built regression-based ML models to predict player value using performance statistics
- Applied feature selection and evaluation techniques to improve prediction accuracy
- Interpreted outputs to extract meaningful insights from data

RESEARCH & AWARDS

California Science & Engineering Fair State Qualifier

- Selected among top **700 of 10,000+** projects statewide
- Recipient of **U.S. Army Award** for excellence in research and innovation

LEADERSHIP & ACTIVITIES

Board Member — OC Engineer (STEAM for All)

- Led outreach initiatives bringing engineering programs to underserved schools; organized events with strong impact

Science Olympiad Competitor and Vice President

- **2nd Place**, California State Tournament (300+ teams); multiple state and regional medals in engineering events

Volunteer — Discovery Cube OC Santa Ana, CA

- Engaged **300–400** daily visitors by explaining scientific exhibits and concepts

INTERESTS

Robotics · AI · Fitness · Healthcare · Sports